Ruoxing (David) Yang

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EDUCATION

Georgetown University August 2022 - May 2026
B.S. Computer Science, Math (Minor), Economics (Minor) GPA 3.91 *Washington, DC*
- Computational Complexity, Differential Privacy, Deep Reinforcement Learning, Approximate Computing
- Computer Science Award - recognized by CS faculty as the best graduating senior.

AWS Certified Cloud Practitioner (CLF-C02) July 2024

WORK EXPERIENCE

Georgetown SecLab (Supervised by Prof. Micah Sherr) March 2024 - May 2026
Research Assistant, Software Developer *Washington, DC*
- Developed CADENCE message-passing simulator: implemented MIRAGE and executed simulation experiments, implemented STEALTH logic, redesigned import and driver logics, introduced new datasets.
- Studied and implemented infrastructure for privacy-preserving push-notification system research.

Georgetown University August 2023 - May 2026
Teaching Assistant *Washington, DC*
- Held office hours and graded projects for Artificial Intelligence, Algorithms, Data Structures, CS 1, C2, and Microeconomics.

RESEARCH PUBLICATIONS

"Mirage: Private, Mobility-Based Routing for Censorship Evasion" March 2024 - August 2025
Network and Distributed System Security Symposium (NDSS) 2026 (Author)
- Helped design MIRAGE routing protocol with Zachary Ratliff, James Mickens, and Micah Sherr. Implemented protocol in CADENCE and handled simulation evaluations.

"DP-Adam-AC" May 2025 – August 2025
Provost Distinguished Fellow
- Studied and modified small language models (SLMs) and bit-quantized models (bitnet) by applying differentially-private fine-tuning using custom DP-Adam-AC optimization (preprint: arXiv:2510.05288).

"A Novel Algorithmic Approach to the Line Addition Problem in Public Transit" May 2024 – August 2024
Lisa J. Raines Fellow
- Designed an algorithm to improve existing public transit networks (preprint: arXiv:2412.12109).

PROJECTS (Code available on GitHub or upon request)

Dynamic Approximate Classification (Computer Vision, Approximate Computing) 2025
- Dynamic vision object classification framework using approximate computing concepts.

Proximal Policy Optimization with Pre-Training (Deep Reinforcement Learning) 2025
- Designed and experimented with PPOPT, which modifies PPO for sparse experience RL (preprint: arXiv:2510.10029)

Video Game Engine (C++, SDL2) 2023
- Created a 2D graphics framework optimized for video-game mechanics.

LEADERSHIP

Provost Distinguished Research Fellow April 2025 – August 2025
- Led research cohort of 10 fellows and hosted weekly research meetings discussing academic research techniques

SKILLS & INTERESTS

- **Programming Languages:** C++/C, Java, Go, Python, SQL, Bash, JavaScript, TypeScript, Stata
- **Technologies:** Linux, Latex, Android, IOS, AWS, SDL2, MySQL, GORM, Git, Caddy, Excel, Adobe Indesign
- **Foreign Languages:** English (Native Proficiency), Chinese (Native Proficiency)
- **Interests:** Running, Musical Theatre, Drums, Guitar.